

Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP397B001-V1

osptek[®]

Revision Record

Rev No.	Date	Description
V.0	2022-05-31	First Release



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1.General Specifications

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	3.97"	inch	-
2	Panel Type 面板类型	IPS	-	-
3	Resolution 分辨率	480RGBx800	Pixel	-
4	Display Mode 显示模式	Normally Black	-	-
5	Number of Colors 颜色数量	16.7M	-	-
6	Viewing Direction 使用视角	ALL	-	Note1
7	NTSC 色彩饱和度	69.5%	-	Typ.
8	Contrast Ratio 对比度	800	-	
9	Luminance 亮度	350	cd/m2	Typ.
10	Module Size 模组尺寸	55.44(H)x96.17(V)x2.1(D)	mm	Note1
11	Panel Active Area 可视区域	51.84x86.40	mm	Note1
12	Pixel Pitch 像素尺寸	0.108x0.108	mm	-
13	Pixel Arrangement 像素排列	RGB-stripe		-
14	Weight 重量		g	-
15	Driver IC 驱动芯片	GC9503CV	-	-
16	Driver IC RAM Size 记忆体	RAM less	bit	-
17	Light Source 背光源	8 LED	-	-
18	Interface 接口方式	MIPI	-	-
19	Operating Temperature 工作温度	-20~+70	°C	-
20	Storage Temperature 存储温度	-30~+80	°C	-

Note 1: Please refer to the mechanical drawing ;

注 1：请参照模组图；

2.Pin Assignments

Pin No. Pin 序号	Symbol 符号	Function 功能描述
1	GND	Ground
2	D0P	MIPI DSI data lane0+
3	D0N	MIPI DSI data lane0-
4	GND	Ground
5	D1P	MIPI DSI data lane1+
6	D1N	MIPI DSI data lane1-
7	GND	Ground
8	CLKP	MIPI DSI clock lane+
9	CLKN	MIPI DSI clock lane-
10	GND	Ground
11	NC	
12	NC	
13	GND	Ground
14	NC	
15	NC	
16	GND	Ground
17	GND	Ground
18	IOVCC	Digit Power Supply for LCM
19	IOVCC	Digit Power Supply for LCM
20	NC	
21	NC	
22	NC	
23	NC	
24	RST	LCM reset signal
25	GND	Ground
26	NC	
27	GND	Ground
28	LEDK	Backlight cathode
29	LEDK	Backlight cathode

30	GND	Ground
31	NC	
32	GND	Ground
33	GND	Ground
34	NC	
35	LEDA	Backlight anode
36	LEDA	Backlight anode
37	GND	Ground
38	VCC	Analog Power Supply for LCM
39	VCC	Analog Power Supply for LCM
40	NC	



3. Electrical Specification

3.1 Absolute Maximum Ratings

Item 项目	Symbol	Value	Unit	Remark
Analog Power Supply Voltage 模拟供电电压	VCI	-0.3~+3.3	V	-
Digital Power Supply Voltage 数字电源电压	VDD	-0.3~+3.3	V	-
I/O Power Supply Voltage I/O端口供电电压	IOVCC	-0.3~3.3	V	-

3.2 Typical Operation Conditions 典型工作条件

Item 项目	Symbol	Min.最小	Typ.典型	Max.最大	Unit
Analog Supply Voltage 模拟供电电压	VCI	2.7	2.8	3.3	V
Digital Supply Voltage 数字电源电压	VDD	2.7	2.8	3.3	V
I/O Supply Voltage 接口电压	IOVCC	1.65	1.8/2.8	3.3	V
Input High Voltage 输入高电平	V_{IH}	$0.8 \cdot IOVCC$	-	IOVCC	V
Input Low Voltage 输入低电平	V_{IL}	0	-	$0.2 \cdot IOVCC$	V
Output High Voltage 输出高电平	V_{OH}	$0.8 \cdot IOVCC$	-	-	V
Output Low Voltage 输出低电平	V_{OL}	-	-	$0.2 \cdot IOVCC$	V

3.3 Backlight Circuit Characteristics 背光功耗

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current 背光电流	I_B	-	40	-	mA
LED Voltage 背光电压	V_f	-	12.8	-	V
Power Consumption 功耗	P_{BL}	-		-	mW

3.4 LCD Current Consumption 液晶面板功耗

Item	Symbol	Typ.	Max.	Unit
Full Mode 正常模式	VCI	-	-	mA
测试条件：VCI=2.8V，IOVCC=2.8V； Interface 驱动类型：行翻转或者列翻转； TN Type=>All Black Pattern. TN型液晶面板=>黑色画面； IPS Type=>All White Pattern. IPS型液晶面板=>白色画面； Temperature：25℃；温度：室温25摄氏度；				
Sleep Mode 休眠模式	VCI	-	-	uA
测试条件：VCI=2.8V，IOVCC=2.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. 除IC内部晶振和面板扫描外，其他功能都暂停工作； Temperature：25℃；温度：室温25摄氏度；				

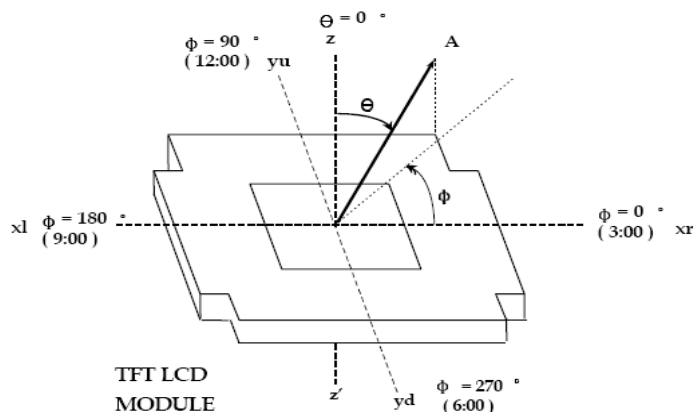
4.Optical Specification

4.1 LCM Optical Characteristics

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Viewing Angle Range 视角	Left	θ_L	$CR \geq 10$	-	85	-	degree
	Right	θ_R		-	85	-	
	Top	θ_T		-	85	-	
	Bottom	θ_B		-	85	-	
Response Time 响应时间		Ton+Toff	$\theta=\phi=0^\circ$	-	35		ms
Contrast Ratio 对比度		CR	$\theta=\phi=0^\circ$	550	800	-	-
Luminance 亮度		L	$\theta=\phi=0^\circ$	-	350	-	cd/m ²
Color Chromaticity (CIE1931) 色坐标	White	W_x	Normal $\theta=\phi=0^\circ$	0.277	0.292	0.307	-
		W_y		0.318	0.333	0.358	
	Red	R_x		0.648	0.663	0.678	
		R_y		0.310	0.325	0.340	
	Green	G_x		0.256	0.271	0.386	
		G_y		0.579	0.594	0.609	
	Blue	B_x		0.119	0.134	0.149	
		B_y		0.107	0.122	0.137	
Uniformity 均匀度		U_L	$\theta=\phi=0^\circ$	80	-	-	%
Flicker 闪烁		-	-	No Visible			-

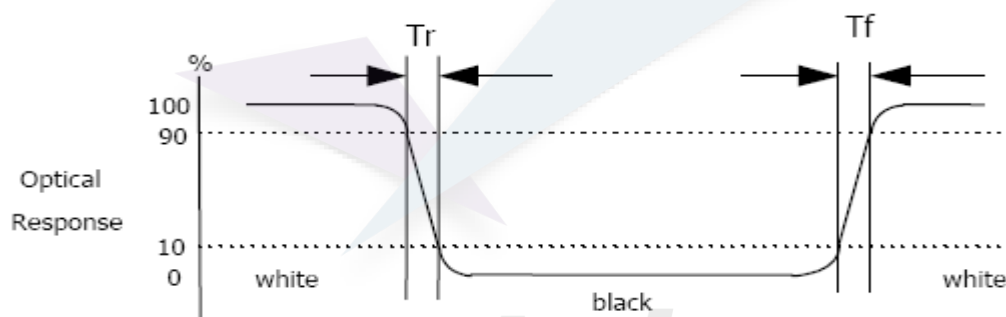
4.2 Measurement system

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, T_r) and from black to white (Falling time, T_f) for additional information.

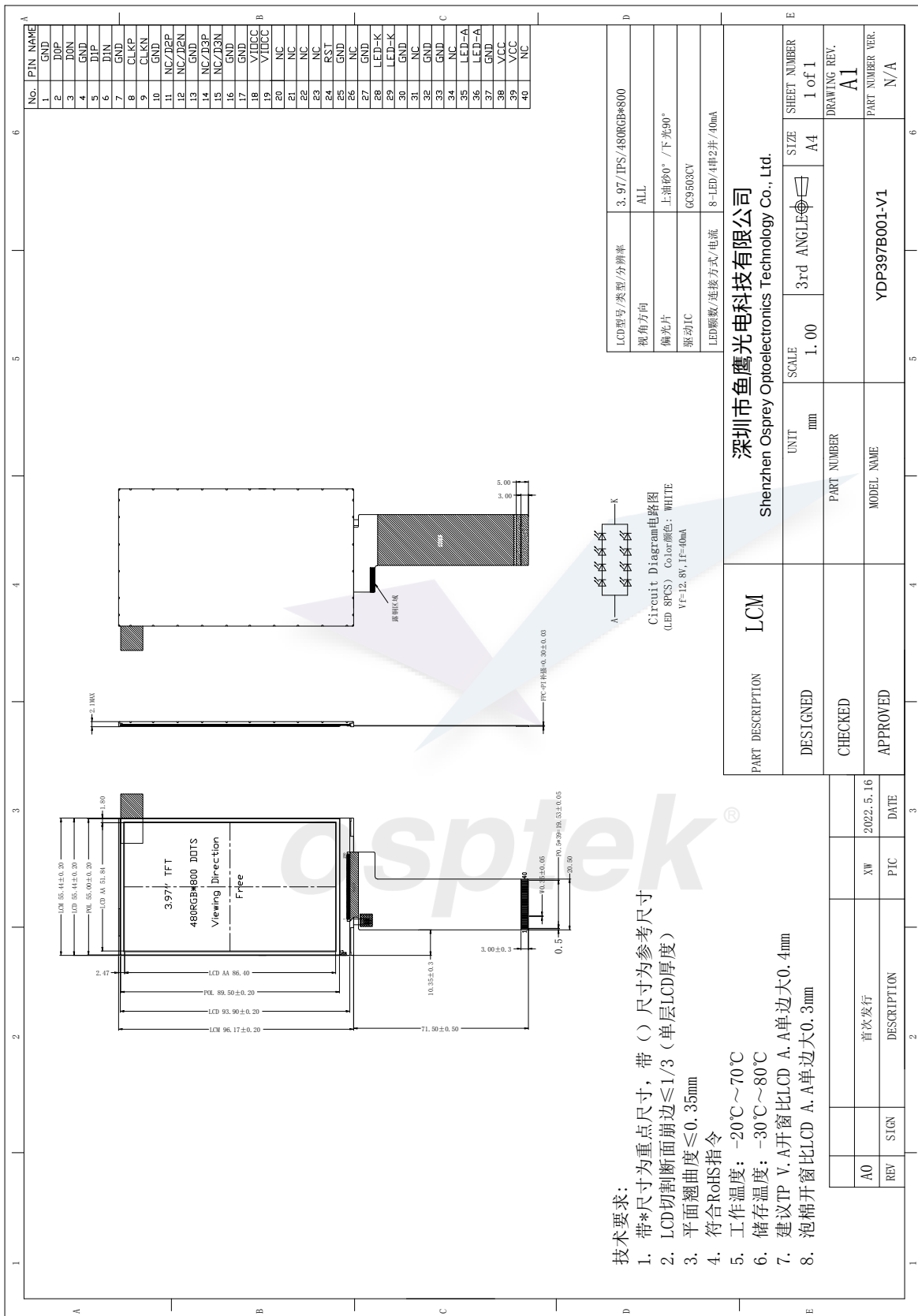
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing



6. Reliability Test Items

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	80±3℃ , 24H ;	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的LCD样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查,样品不允许有以下缺陷: 1.Air bubble in the LCD; 模块中有气泡; 2.Non-display; 不显示; 3.Glass crack; 玻璃破碎; 4. The electrical characteristics requirements shall be satisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-30±3℃ , 24H ;	
High temperature operation 高温运行测试	70±3℃ , 24H ;	
Low temperature operation 低温运行测试	-20±3℃ , 24H ;	
High temperature /humidity 高温高湿	60℃±3℃,90%±3%RH , 24H ;	
Thermal Shock 冷热冲击	-30℃/0.5h~+80℃/0.5h for a total 24 cycles ;	
Vibration Test 振动测试	Frequency 10Hz~55Hz~10Hz Amplitude : 1.5mm, X , Y , Z direction for total 1H ; (Packing condition)	
ESD test 静电测试	±2KV, Human Body Mode, 150pF/330Ω ; ±8KV, Air Mode, 150pF/330Ω ;	

Remark: 注意 :

1. The test samples should be applied to only one test item.

每个被测试的模块只能用于其中的一个测试项目。

2. Sample size for each test item is 2pcs.

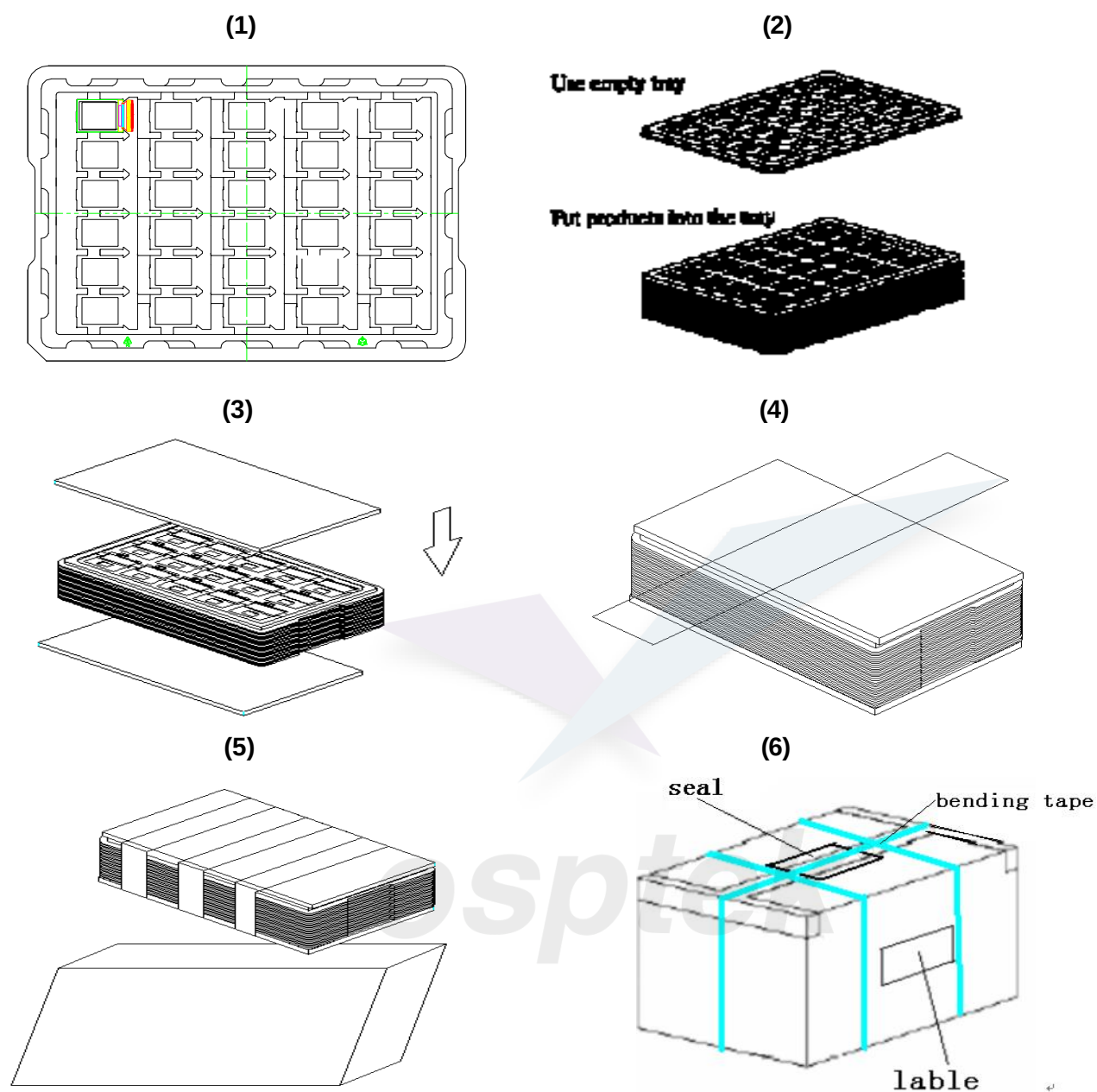
每个测试项目的样品数量为2片。

3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

故障判断标准:基本规格,电气特性,机械特性,光电特性。

7. Packing and Storage Specification(Reference Only)

7.1 Packing Method



1. Put module into tray cavity.
2. Tray stacking.
3. Put 1 foam under the tray stack and 1 foam above.
4. Fix the cardboard to the tray stack with adhesive tape.
5. Put the tray stack into carton.
6. Carton sealing with adhesive tape.

7.2 Storage Method

1. Store in an ambient temperature of $23^{\circ}\text{C}\pm 5^{\circ}\text{C}$, and in a relative humidity of $55\%\pm 15\%$. Don't exceed 12 months and expose to sunlight or fluorescent light.
2. Store in a clean environment, free from dust, active gas, and solvent.
3. Store in antistatic container.



8. Announcements

1. Do not attempt to disassemble or process the LCD module.
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached.
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
4. Other matters in not clear before use, please contact our staff to guide.

